

Customer Churn Prediction

Logistic Regression · Decision Tree · Random Forest

End-to-end ML workflow: EDA, preprocessing, model training, evaluation & prediction

1. Project Overview

This project builds a predictive model to identify customers likely to churn, enabling the business to take proactive retention actions and reduce revenue loss.

Objective

- Predict whether a customer will churn (Yes) or stay (No).
- Identify the most influential factors driving churn.
- Compare three ML models and select the best performer.

Dataset

File: Customer_Churn_Dataset.csv

- Demographics: Age, Gender, City.
- Subscription info: Subscription Type, Monthly Fee, Contract Length.
- Engagement: Login Frequency, Support Tickets.
- Satisfaction: Customer Satisfaction score, Payment Method.
- Target variable: Churn (Yes / No).

2. Workflow Steps

Step 1 — Data Loading & Inspection

The dataset was loaded with Pandas and a full inspection was performed:

- Checked dataset dimensions (df.shape) and available columns.
- Inspected data types and missing values (df.info(), df.isnull().sum()).
- Examined the Churn target variable distribution to detect class imbalance.

Step 2 — Exploratory Data Analysis (EDA)

A comprehensive visual analysis was conducted:

- Count plot showing the distribution of Churn classes.
- Boxplots comparing six numeric features (Age, Monthly Fee, Contract Length, Support Tickets, Login Frequency, Customer Satisfaction) across Churn classes.

Step 3 — Preprocessing

- Dropped the Customer_ID column (non-predictive identifier).
- Encoded the target variable: No -> 0, Yes -> 1.
- Applied OneHotEncoder to categorical columns via ColumnTransformer.
- Split data: 80% training / 20% testing with stratification to preserve class balance.

Step 4 — Model Training & Evaluation

- Built a unified Pipeline per model combining preprocessing and classifier.
- Trained three models: Logistic Regression, Decision Tree, Random Forest.
- Evaluated each model using: Accuracy, Precision, Recall, F1 Score, Confusion Matrix.

Step 5 — Feature Importance & Prediction

- Extracted feature importances from the best-performing Random Forest model.
- Generated a churn prediction for a new customer using real input data.

3. Model Results & Comparison

The table below summarizes the performance of all three models on the test set:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	85.2%	82.1%	79.4%	80.7%
Decision Tree	83.7%	80.5%	78.1%	79.3%
Random Forest ✓ Best	91.4%	89.8%	88.3%	89.0%

Note: ✓ indicates the winning model selected for deployment.

Comparison Analysis

- Random Forest: Clearly outperformed both models with 91.4% accuracy — selected as the final model.
- Logistic Regression: Good, balanced performance at 85.2% despite its simplicity.
- Decision Tree: Weakest performer at 83.7% and most prone to overfitting.

4. Key Factors Influencing Churn

Based on feature importance analysis from the Random Forest model:

Rank	Feature	Relative Importance
1	Customer Satisfaction	Very High ★★★★★
2	Contract Length	Very High ★★★★★
3	Monthly Fee	High ★★★★★☆
4	Support Tickets	Medium ★★★★★☆
5	Login Frequency	Medium ★★★★★☆
6	Subscription Type	Low ★★★★★☆
7	Age	Low ★★★★★☆
8	Payment Method	Low ★★★★★☆

Interpretation

- Customer Satisfaction & Contract Length: The top two drivers. Customers with low satisfaction scores and short-term contracts are most likely to churn.
- Monthly Fee: Higher fees are strongly associated with increased churn probability.
- Support Tickets: Customers opening many support tickets signal a poor experience and higher churn risk.
- Login Frequency: Low engagement correlates with disinterest and potential departure.

5. Recommendations

Based on the project findings, the following actions are recommended:

- **Monitor customer satisfaction scores regularly and launch service improvement initiatives for low-scoring customers.**
- **Offer incentives for customers to renew or upgrade to long-term contracts.**
- **Review pricing strategy for high monthly fee tiers and introduce more flexible plans.**
- **Improve support team efficiency and reduce response times to resolve issues faster.**
- **Deploy the Random Forest model in production for proactive churn scoring and targeted retention campaigns.**